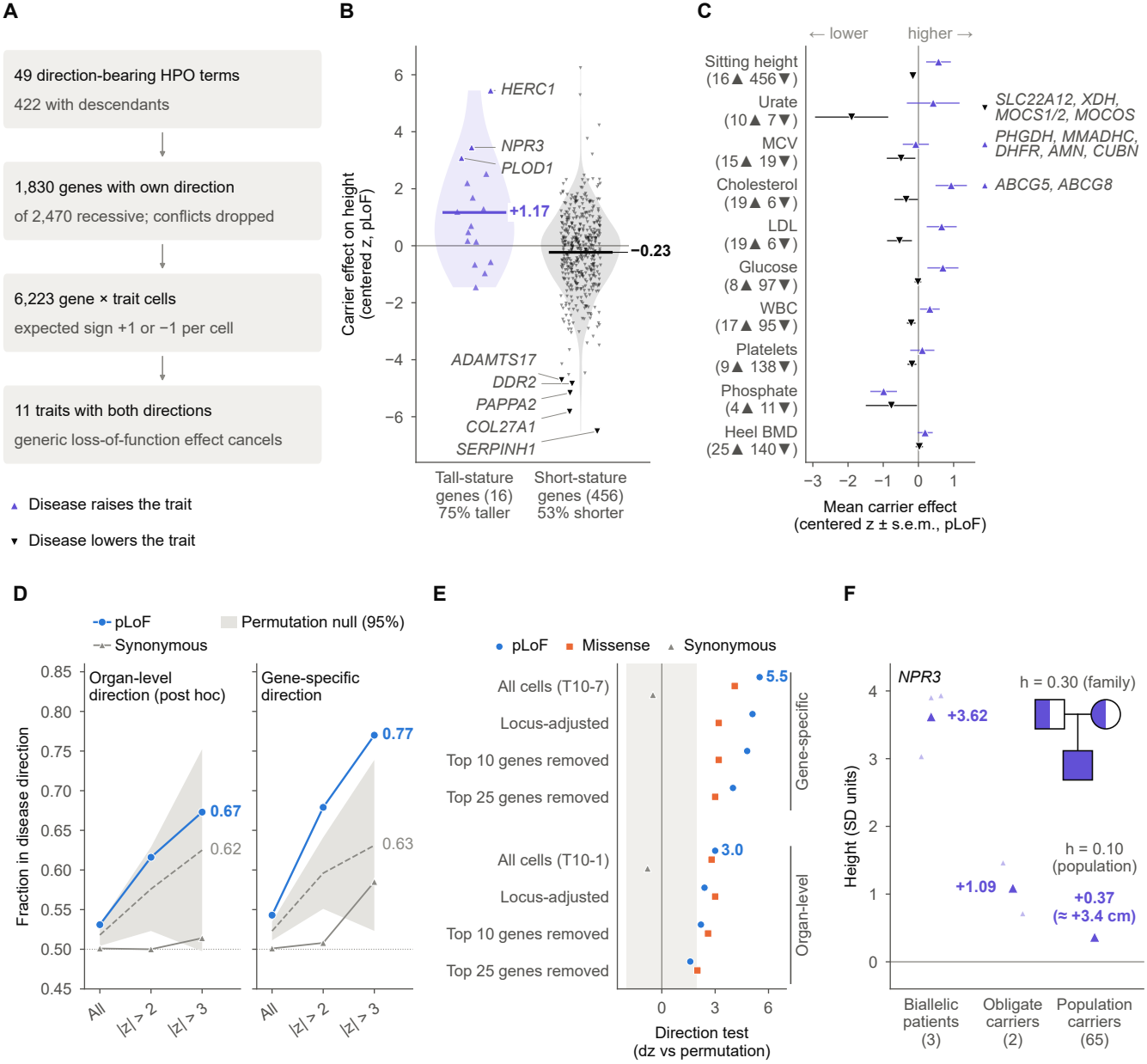

A
49 direction-bearing HPO terms
422 with descendants
1,830 genes with own direction
of 2,470 recessive; conflicts dropped
6,223 gene × trait cells
expected sign +1 or −1 per cell
11 traits with both directions
generic loss-of-function effect cancels
Disease raises the trait
Disease lowers the trait
B
Carrier effect on height (centered z, pLoF)
HERC1
NPR3
PLOD1
+1.17
−0.23
ADAMTS17
DDR2
PAPPA2
COL27A1
SERPINH1
Tall-stature genes (16) 75% taller
Short-stature genes (456) 53% shorter
C
← lower
higher →
Sitting height (16▲ 456▼)
Urate (10▲ 7▼)
MCV (15▲ 19▼)
Cholesterol (19▲ 6▼)
LDL (19▲ 6▼)
Glucose (8▲ 97▼)
WBC (17▲ 95▼)
Platelets (9▲ 138▼)
Phosphate (4▲ 11▼)
Heel BMD (25▲ 140▼)
SLC22A12, XDH, MOCS1/2, MOCOS
PHGDH, MMADHC, DHFR, AMN, CUBN
ABCG5, ABCG8
Mean carrier effect (centered z ± s.e.m., pLoF)
D
pLoF
Permutation null (95%)
Synonymous
Organ-level direction (post hoc)
Gene-specific direction
Fraction in disease direction
0.67
0.62
0.77
0.63
All
|z| > 2
|z| > 3
E
pLoF
Missense
Synonymous
All cells (T10-7)
Locus-adjusted
Top 10 genes removed
Top 25 genes removed
All cells (T10-1)
Locus-adjusted
Top 10 genes removed
Top 25 genes removed
5.5
3.0
Gene-specific
Organ-level
Direction test (dz vs permutation)
F
NPR3
h = 0.30 (family)
+3.62
+1.09
h = 0.10 (population)
+0.37 (≈ +3.4 cm)
Height (SD units)
Biallelic patients (3)
Obligate carriers (2)
Population carriers (65)